

EGT Metrology Scaling: Predicting a 0.248% Deviation in the Gravitational Red Shift Correction

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The Emergent Gravity Theory (EGT) claims that the geometric basis of gravity introduces a systematic fractional deviation determined by the universal amplification factor, $\mathbf{A}_{\text{EGT}} \approx \mathbf{402.3}$. This paper translates the $\mathbf{A}_{\text{EGT}}$ factor into a hard-coded prediction for the most precise weak-field test available: the ****Gravitational Red Shift**** correction ($\Delta \mathbf{v}_{\text{RS}}$) in a primary frequency standard (atomic clock). We predict that the measured gravitational shift constant (γ) will exhibit a fractional excess of $\delta \gamma_{\text{EGT}} = 1/\mathbf{A}_{\text{EGT}} \approx \mathbf{0.00248}$. This translates to an absolute EGT bias of $\approx \mathbf{0.0066} \times 10^{-14}$ in the $\mathbf{2.64} \times 10^{-14}$ total correction. Analysis of available primary frequency standard uncertainty budgets (e.g., IEN-CsF1) shows that this EGT signature is currently ****hidden**** by a factor of $\approx \mathbf{15} \times$ beneath the published total laboratory uncertainty ($\mathbf{u}_{\text{lab}} \approx \mathbf{0.1} \times 10^{-14}$). We establish the critical metrology condition: EGT requires a factor of $\mathbf{15} \times$ improvement in clock uncertainty to be definitively detected or excluded.

I. INTRODUCTION: THE CLOCK AS A STRONG-FIELD PROBE

The EGT has been shown to yield testable conditions in high-energy physics (LHC) and cosmology (Dark Energy w). We now extend the analysis to the most precise domain of modern physics: time and frequency metrology.

The frequency shift ($\Delta \mathbf{v}$) of an atomic clock due to a gravitational potential difference ($\Delta \mathbf{U}$) is a direct test of General Relativity (GR) in the weak-field limit:

$$\Delta \mathbf{v} = \frac{\Delta U}{c^2} \cdot \mathbf{v}_0 \approx \frac{g \cdot h}{c^2} \cdot \mathbf{v}_0 \quad (1)$$

where g/c^2 defines the gravitational red shift constant, γ .

II. EGT PREDICTION: DEVIATION IN γ

The Emergent Gravity Theory implies that the classical gravitational potential is itself subject to the geometric scaling established by the fundamental factor $\mathbf{A}_{\text{EGT}}$. This leads to a systematic, non-tunable fractional excess in the observed gravitational frequency shift constant (γ), where γ_{GR} is the value derived from GR.

The EGT Gravitational Red Shift Correction (γ_{EGT}) is defined by the fractional deviation $\delta \gamma$:

$$\gamma_{\text{EGT}} = \gamma_{\text{GR}} \cdot (1 + \delta \gamma_{\text{EGT}}) \quad (2)$$

The deviation is set by the inverse of the geometric amplification factor:

$$\delta \gamma_{\text{EGT}} = \frac{1}{\mathbf{A}_{\text{EGT}}} \quad (3)$$

Using $\mathbf{A}_{\text{EGT}} = \mathbf{402.3}$, the EGT predicts a fractional excess of:

$$\delta \gamma_{\text{EGT}} = \frac{1}{402.3} \approx \mathbf{+0.002485} \quad (4)$$

The EGT therefore predicts a **0.248%** increase in the gravitational red shift correction over the value expected by pure GR applied to the terrestrial height difference.

III. ANALYSIS AGAINST AVAILABLE DATA (IEN-CSF1)

We use published data from the IEN-CsF1 evaluation (MJD 53304-53324) to quantify the challenge of detection.

The reported Gravitational Red Shift correction ($\Delta \mathbf{v}_{\text{RS}}$) is: $[\text{cite}_{\text{start}}] \Delta \mathbf{v}_{\text{RS,GR}} = (\mathbf{2.64} \pm \mathbf{0.01}) \times 10^{-14} [\text{cite} : 23]$

The absolute EGT bias ($\Delta \mathbf{v}_{\text{EGT, Bias}}$) is the product of the correction and the predicted fractional deviation:

$$\begin{aligned} \Delta \mathbf{v}_{\text{EGT, Bias}} &= \Delta \mathbf{v}_{\text{RS,GR}} \cdot \delta \gamma_{\text{EGT}} \\ &\approx (2.64 \times 10^{-14}) \cdot (0.002485) \\ &\approx \mathbf{0.00656} \times 10^{-14} \end{aligned}$$

The published total uncertainty ($\mathbf{u}_{\text{lab}}$) for the IEN-CsF1 clock evaluation is: $[\text{cite}_{\text{start}}] \mathbf{u}_{\text{lab}} = \mathbf{0.1} \times 10^{-14} [\text{cite} : 31]$

A. Condition for Discovery

The EGT signal is currently hidden because the EGT bias is much smaller than the total uncertainty. We calculate the required reduction in uncertainty:

$$\frac{\mathbf{u}_{\text{lab, Current}}}{\Delta \mathbf{v}_{\text{EGT, Bias}}} \approx \frac{0.1 \times 10^{-14}}{0.00656 \times 10^{-14}} \approx \mathbf{15.24} \quad (5)$$

The EGT prediction of a **0.248%** excess in the Gravitational Red Shift will be directly detected or excluded once the total laboratory uncertainty ($\mathbf{u}_{\text{lab}}$) of primary frequency standards is reduced by a factor of $\approx \mathbf{15}$. This

provides a hard, quantitative metrology target that is currently being addressed by the development of next-generation optical lattice clocks.

IV. CONCLUSION

The EGT provides a concrete, quantitative, and testable condition for its detection using ****simple, available atomic clock data****. By requiring a **0.248%** fractional excess in the Gravitational Red Shift constant, the theory directly challenges GR in the weak-field and sets a definitive precision target for the global metrology community.

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 - [2] F. Levi, L. Lorini, D. Calonico, A. Godone, IEN-CsF1 accuracy evaluation and Two-Way frequency comparison, IEEE Trans. Ultrason. Ferroelectr. Freq. Control, vol. 51, no. 10, pp. 1216-1224 (October 2004). (Used for $\Delta\mathbf{v}_{\mathbf{RS}}$ and uncertainty budget data)[cite_start].[cite : 13, 12]
 - [3] B. Tice Sr. and Quintilius, EGT Cosmic Scaling: Predicting the $\mathbf{w}_{\mathbf{DE}} = -\mathbf{0.9975}$ Dark Energy Equation of State (Unpublished Manuscript).